

Module 87.1: Cognitive Errors vs. Emotional Biases

SCHWESERNOTES - BOOK 4

Traditional finance assumes that individuals act as perfectly rational economic beings who objectively consider all relevant information to make rational decisions, and that this process results in efficient markets. Research results have called these assumptions into question. Kahneman and Tversky pioneered this work in the early 1970s, initially setting logic tests where individuals' intuitive answers were predictably flawed. This supported the idea that human decision-making has systematic biases that lead to irrational decisions.

This work was extended by Kahneman, Tversky, and others, who have suggested that a better understanding of these biases by clients (and by the professionals who work with those clients) should produce securities prices and returns over time that better match the informationally efficient markets of traditional finance theory.

LOS 87.a: Compare and contrast cognitive errors and emotional biases.

Individuals, when facing complex decision-making, often lack the time or ability to derive the optimal course of action prescribed by traditional finance. Cognitive limitations and emotional responses introduce bias into the decision-making process, leading to decisions that are biased (not perfectly rational). Behavioral finance asserts that certain biases, which are not simply random errors, are widespread and therefore predictable.

Cognitive errors are due primarily to faulty reasoning or irrationality. They can arise from not understanding statistical analysis, information processing errors, illogical reasoning, or memory errors. Such errors can possibly be reduced by increased awareness, better training, or more information.

Emotional biases are not related to conscious thought. Rather, they stem from feelings, impulses, or intuition. As such, they are difficult to overcome and may have to be accommodated.

Despite the distinction in grouping biases as either cognitive or emotional, a bias may have elements of both cognition and emotion. When trying to overcome or mitigate biases that are both emotional and cognitive, success is more likely by focusing on the cognitive issues.

LOS 87.b: Discuss commonly recognized behavioral biases and their implications for financial decision making.

Cognitive errors can be divided into **belief perseverance** biases that reflect an irrational reluctance to change prior conclusions and decisions, and **processing errors** where the information analysis is flawed.

Cognitive Errors: Belief Perseverance

Psychologists use the term **cognitive dissonance** to refer to a situation where an individual holds conflicting beliefs or receives information that causes a current belief to be questioned. Cognitive dissonance causes stress that individuals seek to reduce. They may do so by letting go of prior beliefs in favor of the conflicting belief. On the other hand, they might discount the conflicting information or viewpoints by questioning their truth, source, applicability, or significance. To the extent that it is easier to do the latter than the former, bias in favor of currently held beliefs is the result.

- **Conservatism bias** occurs when market participants rationally form an initial view but then fail to change that view as new information becomes available. That is, they overweight their prior probabilities and do not adjust them appropriately as new information becomes available. Individuals displaying this bias tend to maintain prior forecasts and securities allocations, ignoring or failing to recognize the significance of new information. Individuals may react slowly to new data or ignore information that is complex to process.

Example: Conservatism bias

John Molinari allocates assets based on his observation that over the last 80 years, recessions occurred in 20% of those years. When a coworker informs Molinari that the country's central bank has announced a policy change to a tightening of monetary conditions, Molinari does not adjust his recommended asset allocations. Does this reflect conservatism bias?

Answer:

Molinari should consider that the conditional probability of a recession, given that the central bank is tightening, may differ from the unconditional probability of a recession that he previously estimated. He is showing conservatism bias by not considering the impact of this new information.

Conservatism bias may result in market participants holding investments too long because they are unwilling or slow to update a view or forecast. They may be avoiding the mental effort or stress of updating prior beliefs by not considering the implications of new information.

- **Confirmation bias** occurs when market participants focus on or seek information that supports prior beliefs, while avoiding or diminishing the importance of conflicting information or viewpoints. They may distort new information in a way that remains consistent with their prior beliefs.

For example, after buying a car from a given manufacturer, the buyer would exhibit confirmation bias by reading articles about how great cars from that manufacturer are but avoiding news about problems with that particular brand. Because the buyer already purchased the car, information suggesting that it was a bad decision is unwelcome.

Consequences and implications of confirmation bias may include market participants who:

- Consider positive information but ignore negative information.
- Set up a decision process or data screen incorrectly to support a preferred belief.
- Become overconfident about the correctness of a presently held belief.

Market participants can reduce confirmation bias by seeking out contrary views and information—for example, by reading analyses and viewpoints that disagree with a presently held belief, rather than only reading those that reinforce the belief.

- **Representativeness bias** occurs when certain characteristics are used to put an investment in a category and the individual concludes that it will have the characteristics of investments in that category. Individuals systematically make the error of believing that two things that are similar in some respects are more similar in other respects than they actually are.

Two forms of representativeness bias are base-rate neglect and sample-size neglect.

- **Base-rate neglect** refers to analyzing an individual member of a population without adequately considering the probability of a characteristic in that population (the base rate). Consider this example of base-rate neglect: a group was asked to identify the most likely occupation of a man who was characterized as somewhat shy as a salesperson or a librarian. Most participants chose librarian, thinking that most librarians would tend to be more shy on average than salespeople, who tend to be outgoing. Their mistake was in not considering that there are relatively few male librarians and a great number of male salespeople. Even though a greater percentage of librarians may be characterized as somewhat shy, the absolute number of salespeople who could be characterized as somewhat shy is significantly greater.
- **Sample-size neglect** refers to making a classification based on a small and potentially unrealistic data sample. The error is believing the population reflects the characteristics of the small sample.

For example, a fund manager may show strong performance over a three-year time horizon. This may lead investors to assume this is evidence of superior skill. However, examination of longer sample periods shows a lack of persistence; there will be some three-year "winners" even when annual returns results are actually random. The evidence of the lack of persistence of managers' relative performance over time does not support investors' conclusion that this manager will continue to outperform.

Example: Representativeness bias

XYZ company has long been recognized as a growth stock, delivering superior earnings growth and stock price appreciation. While earnings have continued to grow, last year's revenue has not, and neither has the stock price. Under the following two conditions, would an analyst be more likely to buy or sell the stock?

1. The analyst suffers from base-rate and sample-size neglect.
2. The analyst treats the growth classification as representative.

Answer:

If the analyst exhibits sample-size neglect and base-rate neglect biases, the analyst will ignore XYZ's long record as a growth stock, focus on the short-term disappointing results, and may recommend selling the stock without considering the long-term possibility it will revert to growth behavior.

However, if the analyst over-relies on the initial growth classification, the analyst may assume that the stock will return to growth and recommend buying it, without properly considering the reasons for its recent results or their longer-term implications.

Representativeness bias may lead market participants to attach too much importance to a few characteristics based on a small sample size or make decisions based on simple rules and classifications rather than conducting a more-thorough and complex analysis.

- **Illusion of control bias** exists when market participants believe they can control or affect outcomes when they cannot. It is often associated with emotional biases: *illusion of knowledge* (belief you know things you do not know), *self-attribution* (belief you personally caused something to happen), and *overconfidence* (an unwarranted belief that your beliefs will prove to be correct)

An example of control bias can be seen in humans' attempts to control the weather using ritual ceremonies. This allowed them to believe they had some control of the environment, when in fact it is highly unlikely that a ceremony can influence the weather.

Illusion of control may cause market participants to overweight securities for which they believe they have control over outcomes, such as a company they work for or are otherwise associated with. This can result in their portfolios being inadequately diversified.

- **Hindsight bias** is a selective memory of past events, actions, or what was knowable in the past, resulting in an individual's tendency to see things as more predictable than they really are. People tend to remember their correct predictions and forget their incorrect ones. They also overestimate what could have been known. This behavior results from individuals being able to observe outcomes that did occur but not the outcomes that did not materialize. Hindsight bias is sometimes referred to as the *I-knew-it-all-along phenomenon*.

Examples of behavior that reflects hindsight bias include the following:

- Individuals distort their earlier predictions when looking back. This is the tendency to believe that we knew the outcome of an uncertain event all along.
- Individuals tend to view events that have occurred as inevitable.
- Individuals assume they could have foreseen the outcomes of uncertain events.

Hindsight bias can lead to overconfidence in ability to predict outcomes. It may also cause investors to cast aside valid analysis techniques that did not turn out to be correct in favor of poor techniques that turned out well by chance.

Cognitive Errors: Information-Processing Biases

These are related more to the processing of information and less to the decision-making process.

Anchoring and adjustment bias refers to basing expectations on a prior number and overweighting its importance, making adjustments in relation to that number as new information arrives. Examples would be estimating the value of a security relative to its current value or making estimates of earnings per share relative to a previously reported value or relative to a prior estimate. Anchoring leads to underestimating the implications of new information. New data should be considered objectively without regard to any initial anchor point.

Mental accounting bias refers to viewing money in different accounts or from different sources differently when making investment decisions. This conflicts with the idea that security decisions should be made in the context of the investor's overall portfolio of assets based on their financial goals and risk tolerance.

An example of mental accounting is an investor who receives an unexpected bonus at work and chooses to invest it in a very risky biotechnology stock, reasoning that the bonus is "found money" that can acceptably be risked on speculation. In fact, while such a stock may have a place in the investor's portfolio, decisions about whether and how much of it to include should be based on a total portfolio approach. Another example may be an investor who receives an inheritance from a parent and segregates those funds into low-risk bonds out of a desire to "not lose any of the money the parent worked so hard to save." This behavior also conflicts with the principle that investments should be considered in the context of the entire portfolio.

The result of such mental accounting may be that an investor's overall portfolio is not optimal given the investor's circumstances, investment goals, and risk tolerance. It can cause an investor to hold positions that offset each other, rather than considering investments in the context of their correlation of returns. One common form of mental accounting bias is a tendency to view income differently from capital appreciation. This may cause an investor to hold a mix of income-producing and non-income-producing securities that does not match the investor's circumstances.

A study by Camerer et al. (1997)¹ investigated behavior of New York taxi drivers. The study suggests that each new day was a separate account in the minds of the taxi drivers. It appears that New York taxi drivers have a reference rate for daily income and perceive a loss if they fail to meet it (which we may also view as an example of anchoring). On rainy days, demand for taxis is high, while on sunny days, the demand is low. Logically, you would expect taxi drivers to work more hours on rainy days to maximize their incomes. In reality, the opposite was true. Taxi drivers worked longer hours on sunny days as they strove to hit their target income. On rainy days, once the target was achieved, the taxi drivers stopped working.

Framing bias occurs when decisions are affected by the way in which the question or data is "framed." In other words, the way a question is phrased can influence the answer given. Tversky and Kahneman (1980)² illustrate framing bias with the following example.

Example: Framing bias (framing as a gain)

The United States is preparing for the outbreak of an unusual disease, which is expected to kill 600 people. Two alternative programs have been proposed. If Program A is adopted, 200 people will be saved. If Program B is adopted, there is a one-third probability that 600 people will be saved and a two-thirds probability that no one will be saved. Which program will people choose?

Answer:

Program A is typically selected. Although the expected value of both Program A and Program B is 200 lives saved, the majority choice is risk averse. The prospect of saving 200 lives with certainty is more attractive than the risky option with the same expected value.

Example: Framing bias (framing as a loss)

A different group of individuals is given the same issue, but the two programs are framed differently. If Program A is adopted, 400 people will die. If Program B is adopted, there is a one-third probability that nobody will die and a two-thirds probability that 600 will die.

Which program will people choose?

Answer:

In this situation, Program B is typically selected. The majority choice is now risk-taking, with the certain death of 400 people being less acceptable than a two-thirds chance that 600 people will die.

In these examples, the two programs presented (A and B) are identical for both groups, but the framing of the information results in different choices. For the first group, the information was presented in the context of a gain, while for the second group, it was in the context of a loss. Loss aversion—the idea that we fear losses more than we value gains—is then demonstrated by the choices made.

An example of framing bias in an investment context is overestimating the significance of short-term price volatility (risk) without weighing it against long-term considerations. This may result in overly conservative portfolios.

Investment managers must take care to avoid framing bias when creating questions to assess an investor's risk tolerance. Failing to properly assess risk tolerance may identify investors as more or less risk averse than they actually are, resulting in portfolios that are inconsistent with the investors' needs.

Availability bias refers to putting undue emphasis on information that is readily available, easy to recall, or based narrowly on personal experience or knowledge. Availability bias occurs when individuals judge the probability of an event occurring by the ease with which examples and instances come to mind. By the very nature of memory, more-recent events are typically easier to recall than events further in the past, which leads to the bias of attaching too much significance to events that have occurred recently and too little to events that occurred further in the past. People also tend to assume that if something is easily remembered, it must occur with a higher probability.

Example: Availability bias

Imagine a word is picked at random from a dictionary. Is it more likely that the word has the letter *r* as the first letter in the word or the third letter?

Answer:

When faced with this problem, most individuals state that it is more likely the letter *r* will be the first letter. In fact, in the English language, there are approximately three times more words with *r* as the third letter than the first. Individuals find it easier to recall words that begin with *r* than words with *r* as the third letter, which distorts their estimation of probabilities.

Availability bias may lead market participants to choose a manager or investment based on advertising or recalling they have heard the name. They may limit their universe of potential investments to familiar firms, resulting in inappropriate asset allocations and lack of diversification. They may also overreact to recent market conditions while ignoring data on historical market performance, or they may place too much emphasis on events that receive a large amount of media attention.

¹ Colin Camerer, Linda Babcock, George Loewenstein, and Richard Thaler, "Labor Supply of New York City Cab Drivers: One Day at a Time," *Quarterly Journal of Economics* 112 (1997): 407–442.

²Amos Tversky and Daniel Kahneman, "The Framing of Decisions and the Rationality of Choice," Stanford University Department of Psychology, Defense Technical Information Center – Technical Report (1980)

Module 87.2: Emotional Biases

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Professor's Note

Some of the terms we are about to discuss have already come up in the discussion of cognitive biases. In general, if an investor's view is based on unconscious emotion that the holder is unwilling or unable to change, we should regard it as an emotional bias. If a bias can be overcome with a relatively simple change in thought process or information, we should regard it as a cognitive bias.

While there is no formally accepted definition, these six biases generally arise from emotion and feelings rather than through conscious thought:

1. **Loss-aversion bias** arises from feeling more pain from a loss than pleasure from an equal gain. Kahneman and Tversky (1979)¹ investigated differences between how people feel when they gain and when they lose and how that affects behavior when faced with risk. They found that individuals' willingness to take a gamble (risk) was very different when facing a loss or a gain.

Consider the following two scenarios:

Scenario 1: An individual is given \$10. The individual is then given the following options:

- Take an additional \$5 with certainty.
- Flip a coin and win an additional \$10 if it lands heads up or nothing if it lands tails up.

Both options represent a gain relative to the original \$10, and the expected value of the gain is \$5 for either option. Option 1 creates a guaranteed outcome of \$15. Option 2 introduces uncertainty, with equal probabilities

of an outcome of \$10 or \$20. Most individuals chose the riskless Option 1 over the riskier Option 2.

Scenario 2: An individual is given \$20. The individual is then given the following options:

- Take a \$5 loss with certainty.
- Flip a coin and lose nothing if it lands heads up but lose \$10 if it lands tails up.

Both options represent a potential loss relative to the original \$20, and the expected loss is \$5 for either option. Most individuals chose risky Option 2 over the riskless certain loss of Option 1.

In both scenarios, the expected value of the individual's wealth is \$15. The options given to the individuals are also identical, with Option 1 resulting in a guaranteed outcome of \$15 and Option 2 providing equal probabilities of either \$10 or \$20. Yet, when faced with gains, people preferred certainty, and when faced with losses, they preferred risk.

The conclusion is that individuals display asymmetrical responses to gains and losses. Kahneman and Tversky suggested that people look at decisions relative to a reference rate. Anything below the reference rate is seen as a loss, and anything above the reference rate is seen as, a gain. The reference rate in Scenario 1 was the \$10 initially given to the individual, and in Scenario 2, it was \$20. The conflicting responses to the scenarios are explained by attitudes toward gains and losses. Crucially, people fear losses far more than they value gains. Thus, in Scenario 2, they were willing to take the risk in the hope of avoiding a loss.

Professor's Note

Be sure to understand the difference between risk aversion and loss aversion. A risk-averse investor is simply an investor who, given two investments with the same expected returns, would select the investment with the lowest risk. A loss-averse investor is one who feels greater pain (decreases in utility) from losses than satisfaction (increase in utility) from gains. As a result, the individual is more likely to take a risk in the hope of avoiding losses than in the hope of achieving gains.

Consequences of loss-aversion bias may include trading too much by selling for small gains, which increases transaction costs and decreases returns, or incurring too much risk by continuing to hold assets that have deteriorated in quality and lost value. If an initial decline in value occurs, loss-averse investors may take excessive risk in the hope of recovering (investment managers may be particularly susceptible to this behavior). A loss-

averse investor might view a position inappropriately as a gain or a loss based on the framing of the reference point.

2. **Overconfidence bias** occurs when market participants overestimate their own intuitive ability or reasoning. It can show up as *illusion of knowledge* when they think they do a better job of predicting than they actually do. Combined with *self-attribution bias*, individuals may give themselves personal credit when things go right (*self-enhancing*) but blame others or circumstances when things go wrong (*self-protecting*). *Prediction overconfidence* leads individuals to underestimate uncertainty and the standard deviation of their predictions, while *certainty overconfidence* occurs when they overstate the probability they will be right.

While overconfidence is both cognitive and emotional, it is more emotional in nature because it is difficult for most individuals to correct and is rooted in the desire to feel good. Overconfidence bias may cause market participants to underestimate risk, overestimate return, and fail to diversify sufficiently.

3. **Self-control bias** occurs when individuals lack self-discipline and favor short-term satisfaction over long-term goals. Often, individuals are not prepared to make short-term sacrifices to meet their long-term goals. They may favor small payoffs now at the expense of larger payoffs in the future, which is known as *hyperbolic discounting*.

Self-control bias may result in insufficient savings to fund retirement needs, which in turn may cause an investor to take excessive risk to try to compensate for insufficient savings accumulation. It may also result in overemphasis on income-producing assets to meet short-term needs.

Self-control bias might be mitigated by establishing an appropriate investment plan (asset allocation) and a budget to achieve sufficient savings. Both should be reviewed on a regular basis.

4. **Status quo bias** occurs when comfort with an existing situation causes an individual to be resistant to change. If investment choices include the option to maintain existing investments or allocations, or if a choice will happen unless the participant opts out, status quo choices become more likely.

Companies have found that automatically enrolling workers in retirement savings schemes, with the option to opt out, increases participation compared with making it necessary for employees to opt in. Thaler and Sunstein (2008)² argue for framing choices in this way to achieve better participation rates in retirement plans, as well as other choices, such as whether to register as an organ donor.

Consequences of status quo bias may include holding portfolios with inappropriate risk and not considering other, better investment alternatives.

5. **Endowment bias** occurs when an asset is felt to be special and more valuable simply because it is already owned. For example, a spouse may hold on to securities the deceased spouse purchased, for reasons like sentiment that are unrelated to the current merits of the securities. Endowment bias is common with inherited assets and might be detected or mitigated by asking a question such as "Would you make this same investment with new money today?"

In studies, individuals have been asked to state their minimum sales price for an asset they own (say \$25) and the maximum price they would pay now to buy the same asset (say \$23). The price at which they would be willing to sell tended to be higher than the price they would pay. This has been explained as an endowment effect. Once they own an asset, they act as if it is worth more than they would pay.

Market participants who exhibit endowment bias may be failing to sell assets that are no longer appropriate for their investment needs, or they hold assets with which they are familiar because they provide some intangible sense of comfort.

6. **Regret-aversion bias** occurs when market participants do nothing out of excessive fear that actions could be wrong. They attach undue weight to errors of commission (doing something that turns out wrong) and not enough weight to errors of omission (not doing something that would have turned out right). Their sense of regret and pain is stronger for acts of commission. This is quite similar to status quo bias.

Herding behavior is a form of regret aversion where participants go with the consensus or popular opinion. Essentially, participants tell themselves they are not to blame if others are wrong too.

Consequences of regret-aversion bias may include excess conservatism in the portfolio because it is easy to see that riskier assets do at times underperform. Therefore, an investor might not buy riskier assets so as not to experience regret when they decline.

LOS 87.c: Describe how behavioral biases of investors can lead to market characteristics that may not be explained by traditional finance.

In our Equity Investments reading on Market Efficiency, we developed the ideas of anomalies (results that do not fit the prevailing model of securities risks and returns) and market inefficiencies (anomalies that present opportunities to earn positive risk-adjusted returns). Many market anomalies have been explained by small sample size, time period bias, or inadequacies in the specification of prevailing models of returns. In the same way, some anomalies once considered evidence of market inefficiency have been explained by the possible misspecification of risk, leading to inaccurate risk adjustment of returns.

Bubbles and subsequent crashes have been with us as long as trading markets have existed. Some have offered explanations for these extremes of valuation based on rational behavior. Investors who leave markets that they determine are exhibiting characteristics of a bubble may miss out on high returns over extended periods. On the other hand, investors who stay invested based on a belief that they will be able to get out at or near the market top often find that that is not realistic. Either type of mistake, leaving the market too early or staying fully invested too long, can be quite damaging to a fund manager's career. Exploiting the fact that a market is "in a bubble" for an arbitrage profit may not be possible due to the restrictions on and risk of short selling in a rapidly rising market.

While behavioral finance does not provide an overall explanation for bubbles and their aftermath, some have claimed that cognitive and emotional biases exhibited during such periods may have some causal effects. These claims include the following:

- *Overconfidence* may lead to overtrading, underestimation of risk, and lack of diversification.
- Persistently good results combined with *self-attribution* bias can fuel overconfidence, as can *hindsight* bias (as investors give themselves credit for choosing profitable stocks in a bull market).
- *Confirmation bias* may lead investors to ignore or misinterpret new information suggesting that valuations will not continue to rise, or to misinterpret initial decreases in asset values as simply another buying opportunity.
- *Anchoring* may cause investors to believe recent highs are rational prices even after prices begin their eventual decline.
- Fear of *regret* may keep even very skeptical investors in the market.

One anomaly that has been persistent over time is the value/growth anomaly. Value stocks (low market-book, low P/E, high dividend yield stocks) have outperformed growth stocks (high P/E, low dividend yield stocks) over long periods. Fama and French (1992) found that adding risk factors for firm size and book-to-market ratio to a model of stock returns eliminated the evidence of value stock outperformance. These results suggest that the extra returns to value stocks were compensation for additional risks captured by firm size and book-to-market ratios.

Others have suggested behavioral factors as the cause of the seeming outperformance of value stocks. The **halo effect** is a version of representativeness in which a company's good characteristics, such as fast growth and a rising stock price, are extended into a conclusion that it is a good stock to own, leading to overvaluation of growth stocks.

The fact that investors tend to invest heavily in firms in their domestic country in a global portfolio, or more heavily in firms operating in their region of a country, is considered anomalous in that rationality suggests greater diversification. Such **home bias**, it is claimed, may result from a belief that they have better access to information or simply an emotional desire to invest in companies "closer to home." Similarly, investors may underestimate the risk or overestimate the future returns of firms whose products they use or firms for which they are exposed to a great amount of positive marketing messages.

¹Daniel Kahneman and Amos Tversky, "Prospect Theory: An Analysis of Decision Under Risk, *Econometrica* 47, no.2 (1979): 263–91.

²Richard H. Thaler and Cass R. Sunstein, *Nudge: Improving Decisions About Health, Wealth, and Happiness* (New York: Penguin Group, 2008).

Reading 87: Key Concepts

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LOS 87.a

Cognitive errors result from the inability to analyze information or from basing decisions on partial information. Individuals try to process information and make rational decisions, but they may lack the capacity or sufficient information to do so. Cognitive errors can be divided into belief perseverance errors and processing errors.

Emotional biases are caused by the way individuals frame information and decisions, rather than the process used to analyze and interpret information. Emotional bias is more of a spontaneous reaction.

LOS 87.b

Cognitive Errors: Belief Perseverance

- Conservatism bias.
- Confirmation bias.
- Representativeness bias.
- Control bias.
- Hindsight bias.

Cognitive Errors: Information Processing

- Anchoring and adjustment.
- Mental accounting bias.

- Framing bias.
- Availability bias.

Emotional Biases

- Loss-aversion bias.
- Overconfidence bias.
- Self-control bias.
- Status quo bias.
- Endowment bias.
- Regret-aversion bias.

LOS 87.c

Many reported anomalies have been explained by inadequate specification of security risk, but some have attempted to explain the existence of persistent anomalies as the results of cognitive errors and emotional biases.

Behavioral finance has not explained bubbles and crashes, but some cognitive errors and emotional biases may be exhibited during bubbles and crashes.